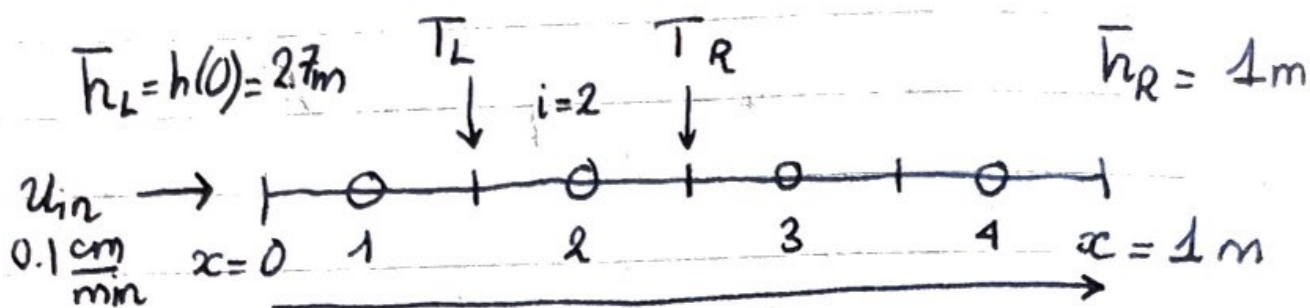


2. Discretize.



$$\Delta x = \delta x = 1/4 = 0.25\text{m}$$

$$i=2,3 \quad \nabla \cdot u = 0 \Rightarrow -Q_L + Q_R = 0 \quad (*)$$

$$Q_L = \underbrace{T_L}_{\frac{2}{\delta x} \left(\frac{1}{K_L} + \frac{1}{K_i} \right)^{-1}} (h_L - h_i), \quad Q_R = \underbrace{T_R}_{\frac{2}{\delta x} \left(\frac{1}{K_i} - \frac{1}{K_R} \right)^{-1}} (h_i - h_R)$$

Substitute in (*):

$$\begin{aligned} \therefore -T_L (h_L - h_i) + T_R (h_i - h_R) &= 0 \\ \Rightarrow -T_L h_L + (T_L + T_R) h_i - T_R h_R &= 0 \end{aligned}$$

$$i=1 \text{ (leftmost cell)} \quad -Q_L + Q_R = 0$$

$$Q_L = \underbrace{\tilde{T}_L}_{\frac{2}{\delta x} \cdot K_1} (\bar{h}_L - h_1), \quad Q_R = \underbrace{T_R}_{\frac{2}{\delta x} \left(\frac{1}{K_i} - \frac{1}{K_R} \right)^{-1}} (h_1 - h_R)$$

$$\begin{aligned} Q_L &= u_{in} \\ \therefore -\tilde{T}_L (\bar{h}_L - h_1) + T_R (h_1 - h_R) &= 0 \\ \Rightarrow -\tilde{T}_L \bar{h}_L + \sum_1 h_1 - T_R h_R &= 0 \end{aligned}$$

$$i=4 \text{ (Rightmost cell)} \quad -Q_L + Q_R = 0$$

$$Q_L = \frac{T_L}{\frac{2}{\delta x} \left(\frac{1}{K_L} + \frac{1}{K_4} \right)^{-1}} (h_{L3} - h_3), \quad Q_R = \frac{\tilde{T}_{R4}}{\frac{2}{\delta x} \left(\frac{1}{K_4} + 1 \right)^{-1}} (h_4 - \bar{h}_R)$$

$$\therefore -T_{L4} h_3 + \sum_4 h_4 - \tilde{T}_{R4} \bar{h}_R = 0$$

$$-T_{L4} (h_{L3} - h_4) + \tilde{T}_{R4} (h_4 - \bar{h}_R) = 0$$

$$\Rightarrow -T_{L4} h_{L3} + \sum_4 h_4 - \tilde{T}_{R4} \bar{h}_R = 0$$

Since we know: $\bar{h}_L = h(0) \approx 2.7 \text{ m}$.

$\bar{h}_R = h(1) \approx 1 \text{ m}$.

$u_{in} = 0.1 \frac{\text{cm}}{\text{min}} \times 1.67 \times 10^{-5} \frac{\text{m}}{\text{s}}$

$$(i=1) \Rightarrow -\tilde{T}_L (2.7) + \sum_1 h_1 - T_{R1} h_{R1} = 0$$

$$(i=4) \Rightarrow -T_{L4} h_{L3} + \sum_4 h_4 - \tilde{T}_{R4} (1) = 0.$$

$$\begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} \begin{bmatrix} \sum_1 & -T_{R1} & 0 & 0 \\ -T_{L2} & \sum_2 & -T_{R2} & 0 \\ 0 & -T_{L3} & \sum_3 & -T_{R3} \\ 0 & 0 & -T_{L4} & \sum_4 \end{bmatrix} \begin{bmatrix} h_1 \\ h_2 \\ h_3 \\ h_4 \end{bmatrix} = \begin{bmatrix} \tilde{T}_L (2.7) \\ 0 \\ 0 \\ \tilde{T}_{R4} \end{bmatrix}$$

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